

VECTOR SPACE

Complete Study Guide + Exam Questions

Topics Covered: Definition · 10 Axioms · Examples · Theorems · Subspaces · Intersection

1. What is a Vector Space?

A **Vector Space** V is a non-empty set of objects (called vectors) — these can be actual vectors, matrices, functions, etc. — on which two operations are defined:

- **Vector Addition:** Adding two elements of V gives another element of V .
- **Scalar Multiplication:** Multiplying an element of V by a scalar (from a field F) gives another element of V .

The **Field F** is the set of scalars — usually real numbers $\mathbb{R}$ or complex numbers $\mathbb{C}$.

Easy Way to Remember: Think of V like a club. A vector space means: if you do any allowed operation on members of the club, the result must still be a club member.

2. The 10 Axioms (Properties)

For V to be a vector space, ALL 10 properties must hold for all a, b, c in V and all α, β in F :

#	Property	Formula	Meaning
1	Closure under +	$a+b \in V$	Sum stays inside V
2	Commutativity	$a+b = b+a$	Order of addition does not matter
3	Associativity	$a+(b+c) = (a+b)+c$	Grouping does not matter
4	Zero Element	$\exists 0 \in V: a+0 = a$	Zero vector acts as identity
5	Additive Inverse	$\exists -a: a+(-a) = 0$	Every vector has a negative
6	Closure under $\times$	$\alpha a \in V$	Scalar $\times$ vector stays in V
7	Associativity $\times$	$\alpha(\beta a) = (\alpha\beta)a$	Scalars combine nicely
8	Distributive 1	$\alpha(a+b) = \alpha a + \alpha b$	Scalar distributes over vector sum
9	Distributive 2	$(\alpha+\beta)a = \alpha a + \beta a$	Sum of scalars distributes
10	Identity Scalar	$1 \cdot a = a$	Multiplying by 1 changes nothing

★ *Note: Properties 1 and 6 are called CLOSURE properties — the most important ones to check first!*

★ *Note: If even ONE property fails $\rightarrow V$ is NOT a vector space. You only need to show ONE failure!*

3. Types of Vector Spaces

- **Real Vector Space:** Scalar field $F = \mathbb{R}$ (real numbers). Example: $\mathbb{R}^n$.

- **Complex Vector Space:** Scalar field $F = \mathbb{C}$ (complex numbers). Example: set of all complex numbers over $\mathbb{R}$.

4. Standard Examples (IS a Vector Space)

(i) $\mathbb{R}^n$ — Ordered n -tuples of real numbers

$$V = \{(a_1, a_2, \dots, a_n) : a_i \in \mathbb{R}\}$$

$$\text{Addition: } (a_1, \dots, a_n) + (b_1, \dots, b_n) = (a_1 + b_1, \dots, a_n + b_n)$$

$$\text{Scalar Mult: } \alpha(a_1, \dots, a_n) = (\alpha a_1, \dots, \alpha a_n)$$

All 10 conditions are satisfied $\rightarrow V = \mathbb{R}^n$ is a real vector space. Zero element = $(0, 0, \dots, 0)$.

(ii) Set of all Complex Numbers $\mathbb{C}$ over $\mathbb{R}$

$$V = \{a + ib : a, b \in \mathbb{R}, i = \sqrt{-1}\}$$

$$\text{Addition: } (a + ib) + (c + id) = (a + c) + i(b + d)$$

$$\text{Scalar Mult: } c(a + ib) = ca + i(cb), c \in \mathbb{R}$$

All 10 conditions satisfied $\rightarrow \mathbb{C}$ over $\mathbb{R}$ is a vector space.

(iii) Set of all $m \times n$ Matrices over $\mathbb{R}$

V = set of all $m \times n$ matrices with real entries.

$$\text{Addition: } (a_{ij})_{m \times n} + (b_{ij})_{m \times n} = (a_{ij} + b_{ij})_{m \times n}$$

$$\text{Scalar Mult: } c(a_{ij})_{m \times n} = (ca_{ij})_{m \times n}, c \in \mathbb{R}$$

All 10 conditions satisfied $\rightarrow V$ is a real vector space.

5. Non-Examples (NOT a Vector Space)

Polynomials of EXACTLY degree n

V = set of all polynomials of degree exactly n . Take $P_n = x^n + \dots$ and $Q_n = -x^n + \dots$. Their sum $P_n + Q_n$ has degree $(n-1)$, which is NOT in V . Closure property fails $\rightarrow V$ is NOT a vector space.

Ordered pairs with non-standard operations

V = set of ordered pairs (x, y) of real numbers with:

$$\text{Addition: } (x_1, y_1) + (x_2, y_2) = (2x_1 - 3x_2, y_1 - y_2)$$

$$\text{Scalar Mult: } \alpha(x_1, y_1) = (\alpha x_1/3, \alpha y_1/3)$$

Check commutativity (Property 2):

$$a + b = (2x_1 - 3x_2, y_1 - y_2) \text{ but } b + a = (2x_2 - 3x_1, y_2 - y_1)$$

These are NOT equal $\rightarrow$ Property 2 fails $\rightarrow V$ is NOT a vector space.

★ Note: You only need to show any ONE failure to conclude it is not a vector space.

6. Important Theorems on Vector Spaces

Let V be a vector space over F . Then:

- $k \cdot 0 = 0$ — Any scalar times the zero vector = zero vector.
- $0 \cdot u = 0$ — The zero scalar times any vector = zero vector.
- $ku = 0 \iff k = 0 \text{ OR } u = 0$ — Product is zero only if at least one factor is zero.
- $(-k)u = k(-u) = -(ku)$ — Negative distributes naturally.

★ *Note: These theorems are frequently asked in exams — memorize all 4!*

7. Subspace

A non-empty subset W of a vector space V is called a **Subspace** of V if W is itself a vector space over the same field F with the same operations.

Subspace Test Theorem (Only 2 conditions needed!)

W is a subspace of V if and only if:

- **Condition (a):** The zero vector 0 belongs to W .
- **Condition (b):** For every $u, v \in W$ and $\alpha, \beta \in F \rightarrow \alpha u + \beta v \in W$ (closed under linear combination).

★ *Note: You do NOT need to check all 10 axioms! Just check these 2 conditions.*

Special Subspaces

- **Improper Subspace:** V itself is a subspace of V .
- **Trivial Subspace:** $\{0\}$ — the set containing only the zero vector.

8. Subspace Examples

Example 1 (IS a Subspace): $S = \{(x, y, z) \in \mathbb{R}^3 : y=0, z=0\}$

- $(0, 0, 0) \in S$ ✓
- Take $u=(x_1, 0, 0)$, $v=(x_2, 0, 0)$. Then $\alpha u + \beta v = (\alpha x_1 + \beta x_2, 0, 0) \in S$ ✓

Both conditions hold $\rightarrow S$ is a subspace of $\mathbb{R}^3$.

Example 2 (NOT a Subspace): $S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = z^2\}$

Take $a=(3, -4, 5)$ and $b=(-3, 4, 5)$. Both satisfy $x^2 + y^2 = z^2$.

But $a+b = (0, 0, 10)$ and $0^2 + 0^2 \neq 10^2 \rightarrow a+b \notin S$. Closure fails $\rightarrow$ NOT a subspace.

9. Intersection of Subspaces

Theorem: The intersection of two (or any family of) subspaces of V is also a subspace of V .

★ *Note: The UNION of two subspaces is generally NOT a subspace.*

10. EXAM QUESTIONS

Short Answer / 2 Marks

Q1.

Define a vector space over a field F .

Q2.

State the 10 axioms of a vector space.

Q3.

What are closure properties in a vector space?

Q4.

Define a subspace. How is it different from a vector space?

Q5.

State the subspace test theorem.

Q6.

What are trivial and improper subspaces?

Q7.

If $ku=0$ in a vector space, what can you conclude?

Q8.

Is the intersection of two subspaces always a subspace? What about union?

Medium Answer / 4-5 Marks

Q9.

Show that $\mathbb{R}^n$ (set of all ordered n -tuples) is a real vector space.

Q10.

Show that the set of all $m \times n$ matrices over $\mathbb{R}$ forms a real vector space.

Q11.

Show that the set of complex numbers $\mathbb{C}$ forms a vector space over the field $\mathbb{R}$.

Q12.

Show that the set of all polynomials of degree EXACTLY n is NOT a vector space.

Q13.

Prove that for any scalar $k \in F$ and $0 \in V$, $k \cdot 0 = 0$.

Q14.

Prove that $0 \cdot u = 0$ for any $u \in V$.

Long Answer / Example-Based

Q15.

Let V be the set of ordered pairs (x,y) with addition defined as $(x_1,y_1)+(x_2,y_2) = (2x_1-3x_2, y_1-y_2)$ and scalar multiplication as $\alpha(x_1,y_1) = (\alpha x_1/3, \alpha y_1/3)$. Show that V is NOT a vector space.

Q16.

Let $S = \{(x,y,z) \in \mathbb{R}^3 : y=z=0\}$. Show that S is a subspace of $\mathbb{R}^3$.

Q17.

Let $S = \{(x,y,z) \in \mathbb{R}^3 : x^2+y^2=z^2\}$. Show that S is NOT a subspace of $\mathbb{R}^3$.

Q18.

Prove that the intersection of two subspaces S and T of a vector space V is a subspace of V .

Q19.

Give an example to show that the union of two subspaces is NOT necessarily a subspace.

Best of luck in your exam! — Prof. Claude